Fire-See

A free, public dashboard that estimates wildfire risk for the forests of Ourense, Galicia, by stitching together open satellite, weather and geospatial datasets. Built as a research-grade prototype that anyone — rangers, journalists, residents — can read, share and learn from.

ZONES TRACKED	8 forest perimeters in Ourense province
LIVE DATA SOURCES	9 (all open and free)
MODEL AUC	0.84 (up from 0.74 in V1)
REFRESH RATE	Weather and risk every 5 minutes
LIVE URL	perspectives-on-ai-and-sustainabili.vercel.app

The problem we are tackling

Galicia loses tens of thousands of hectares of forest every fire season. Local rangers, civil-protection workers and residents need a single place to see **which forest zones are most at risk right now**, why, and what conditions are coming over the next week. Existing official tools are either continent-wide (so the resolution is too coarse for a Galician valley) or behind paywalls. Fire-See fills that gap with a single dashboard that ranks the eight key forest zones in Ourense province on a 0–100 % risk scale, refreshed live.

It is explicitly **not** an emergency service — for live emergencies people still call 112 — but it is a public tool for awareness, preparation and journalism.

How it works in thirty seconds

Fire-See pulls live data from nine open sources, runs each forest zone through a machine-learning model trained on five fire seasons of real Galician fires, and shows the result on a 3-D map. The whole pipeline is open source, deployed on Vercel, and uses only free APIs.

The nine data sources

Source	What it gives us	Refresh
AEMET	Live weather (station 1690A · Ourense)	5 min
NASA FIRMS · VIIRS	Where satellites see active fires	10 min
Open-Meteo	7-day weather forecast (FWI inputs)	1 h
Copernicus EFFIS	EU fire-danger raster (map overlay)	Daily
NASA GIBS	Satellite imagery tiles	Daily
OpenStreetMap	Fire stations / hospitals / water points	On demand
OpenRouteService	Drive-time evacuation circles	On demand

Sentinel-2 (GEE)	Vegetation health (NDVI) and moisture (NDMI)	Weekly

The model: a Random Forest of 200 judges

The risk number for each zone comes from a **Random Forest classifier** with 200 decision trees. Picture 200 small judges, each one looking at the same forest patch and voting "risky" or "safe" based on a handful of clues. We average their votes — the percentage of judges that voted "risky" is the headline score on the dashboard.

Why a Random Forest? It is the standard choice for tabular classification at our data scale (thousands of examples, dozens of features). It handles non-linear effects out of the box, does not need feature scaling, gives us feature-importance scores for free, and we can export the trees as a small JSON file that runs in the browser without any heavy runtime dependency. We tried gradient boosting (HistGradientBoosting) and stacking ensembles too — Random Forest stayed competitive while keeping the deployment trivial.

What each judge weighs

Clue	What it captures	Weight
Elevation	How high above sea level the patch is	19.8%
Surface temperature	How hot the ground is right now	18.1%
Distance to urban areas	How remote the patch is	16.4%
Slope steepness	How much fire could race uphill	12.4%
Vegetation moisture (NDMI)	How wet the plants are	11.9%
Vegetation greenness (NDVI)	How healthy the canopy looks	11.9%
Slope aspect	Which compass direction the slope faces	9.5%

Each clue is a column in the model's input table. Together they describe the "personality" of a forest patch on the day in question.

Training data

We taught the judges by showing them **2,929 real examples** from Galician fire seasons 2018–2022. Half were patches that actually burned (NASA's satellite-derived MCD64A1 burn map flagged them); the other half were similar-looking patches on forest / shrub / grass land that did not burn that year. The model learns the patterns that show up *before* a patch catches fire.

Where the original methodology comes from

Our first model (V1) was a direct Galician adaptation of **Piao et al. (2022)**, "Forest fire susceptibility assessment using Google Earth Engine in Gangwon-do, Republic of Korea" (*Geomatics, Natural Hazards and Risk*). The paper trains a Random Forest on a stack of Sentinel-2 vegetation indices, MODIS land-surface temperature and SRTM terrain features, all extracted through Google Earth Engine — exactly the recipe we ported to Ourense. Their reported AUC on a Korean test set was **0.835**; our V1 reached 0.74 on Galician data, and V3 (described next) closes that gap to 0.84.

Improving the model: V1 → V3

Our first model (V1, deployed early 2026) hit an AUC of 0.74 — decent but with several quiet flaws. V3 (May 2026) rebuilds the dataset and the pipeline from scratch and lifts AUC to 0.84.

What changed

- **Re-extracted the training data from Google Earth Engine.** Instead of using one summer-median snapshot per forest patch, we now take a 30-day window of NDVI / NDMI / surface temperature *strictly before* each fire date. This stops the model from seeing post-burn imagery and learning the wrong thing.
- **Paired the burned and non-burned classes on date.** Each non-burned example inherits its date from a randomly-chosen burned example. That guarantees both classes share the same calendar distribution, removing the seasonal leak (without it the model would trivially separate classes via "is it summer?").
- **Restricted sampling to land that can actually burn.** ESA WorldCover labels every pixel of Europe as urban, water, crops, tree-cover, shrubland, etc. We now sample only the tree-cover, shrubland and grassland classes — so the model is not separating "forest vs. river" but rather "forest that burned vs. forest that did not."
- **Dropped the broken "distance to roads" feature.** The original feature pulled from a global roads dataset that returns a constant value in our bbox, so it carried zero signal. Replaced it with "distance to urban areas" using ESA WorldCover, which actually varies.
- **More years of data — five fire seasons instead of one.** Training samples grew from 1,200 to 2,929. We also moved from a single 70/30 split to **5-fold cross-validation**, which gives a more honest estimate of how well the model generalises.
- **Tested ensembling — kept it simple.** HistGradientBoosting and stacking ensembles bumped AUC by another 0.5–1 point, but the gain was not worth the deployment complexity of new tree formats in the browser. We stuck with Random Forest.

Side-by-side metrics

	V1 (before)	V3 (after)	Change
AUC (5-fold CV)	0.74	0.84	+0.10
Accuracy	67 %	79 %	+12 pp
Cohen's κ	0.33	0.48	+0.15
Training samples	1 200	2 929	+1 729
Fire seasons covered	1 (median 2017–2023)	5 (2018–2022)	—
Trees in the forest	200	200	—

An honest note. The 0.84 above is from random 5-fold cross-validation (the same metric V1 was reported on). If we use the harder "leave one whole year out" test — train on four fire seasons and predict a year the model has never seen — AUC drops to around 0.66. That gap is the gap between "sees similar weather" and "sees a new climate." We report both on the methodology page so users can read the score with the right caveats.

What ships on the dashboard

- **3-D map view.** Eight forest zones extruded by current risk score, plus optional satellite, EU FWI and historical-fire overlays.
- **Zone deep-dive pages.** Permalink per zone with a hero risk gauge, weather tiles, NDVI / NDMI sparklines and a 7 / 30 / 90-day risk history.
- **Fire Weather Index (FWI).** Computed in your browser using Van Wagner's 1987 equations — the same standard the EU uses for all its official fire warnings.
- **Drought, air-quality and incidents pages.** Side panels for KBDI / SPI drought signals, PM2.5 smoke from Copernicus, and a chronological incidents timeline clustered from satellite hotspots.
- **Hotspot context panel.** Click any satellite-detected fire on the map to see the nearest fire stations, hospitals, water points and 10 / 15-minute drive-time circles.
- **Embed widgets.** Council websites or news outlets can drop a live risk badge for any of the eight zones with one line of HTML.
- **Open data, RSS and iCal.** Every dataset is downloadable as CSV or GeoJSON. There is an RSS feed for high-risk zones and an iCal feed for upcoming Very-High FWI days.
- **Plain-language methodology and glossary.** Every technical term has a click-to-explain tooltip linked to a permalinked entry.

Stack

Next.js 15 · React 19 · Mapbox GL · Tailwind CSS · Upstash Redis (KV) · Vercel · Python (scikit-learn) and Google Earth Engine for training.

Where to look

LIVE DASHBOARD	perspectives-on-ai-and-sustainabili.vercel.app
SOURCE CODE	github.com/pablo-munoz/perspectives-on-ai-and-sustainability
METHODOLOGY PAGE	/methodology
OPEN DATA AND API DOCS	/data
AUTHOR	Pablo Muñoz

References

Piao, Y. et al. (2022). *Forest fire susceptibility assessment using Google Earth Engine in Gangwon-do, Republic of Korea.* Geomatics, Natural Hazards and Risk, 13(1), 432–450.
Van Wagner, C. E. (1987). *Development and Structure of the Canadian Forest Fire Weather Index System.* Forestry Technical Report 35, Canadian Forestry Service.